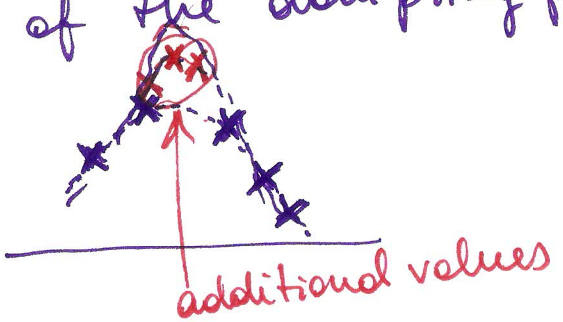
Bode Plot is log.

⇒ to have suitably distributed frequencies $w_1, w_2 \dots\dots$!
ATTENTION.

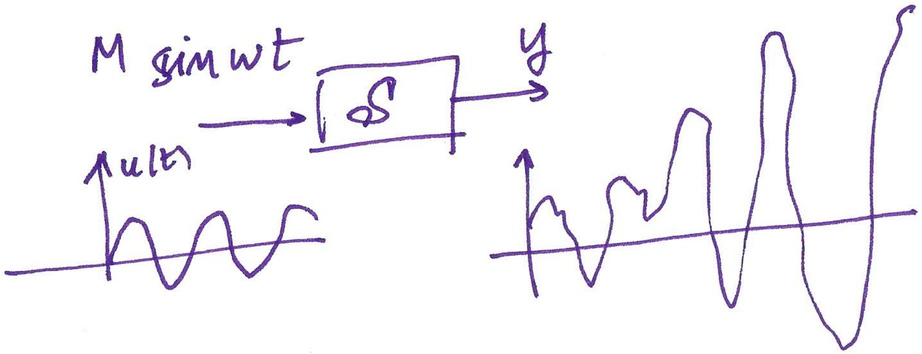
Possibly we can "refine" the plot in some zone.

EX: Where couple of poles are present ⇒ to understand the magnitude of the damping parameter ⇒ more frequencies



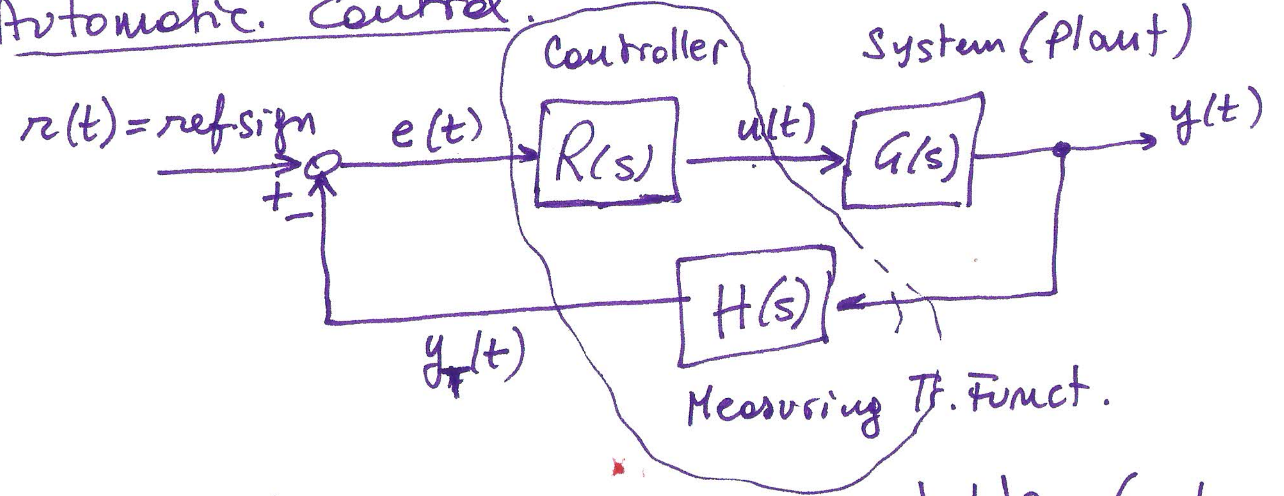
What to do if the system is NOT asymptotically stable.

⇒ open loop control of the system is not suitable.
(output can diverge!)



We could think to "stabilize" the system.

From Automatic control.



If $R(s)$, $H(s)$ s.t. the loop is stable (closed loop stability)

$\Rightarrow$ $r(t) = M_r \sin(\omega t)$ $\Rightarrow$ $y_p(t) = M_r \cdot |T_{yr}(j\omega)| \cdot \sin\left(\omega t + \measuredangle T_{yr}(j\omega)\right)$

perm. part

$u_p(t) = M_r\, |T_{ur}(j\omega)| \sin\left(\omega t + \measuredangle T_{ur}(j\omega)\right)$

$\Rightarrow$ We can measure $\dfrac{|T_{yr}(j\omega)|}{|T_{ur}(j\omega)|} \Rightarrow |G(j\omega)|$

from comparison between $y_p(t)$ and $u_p(t) \rightarrow \measuredangle G(j\omega)$

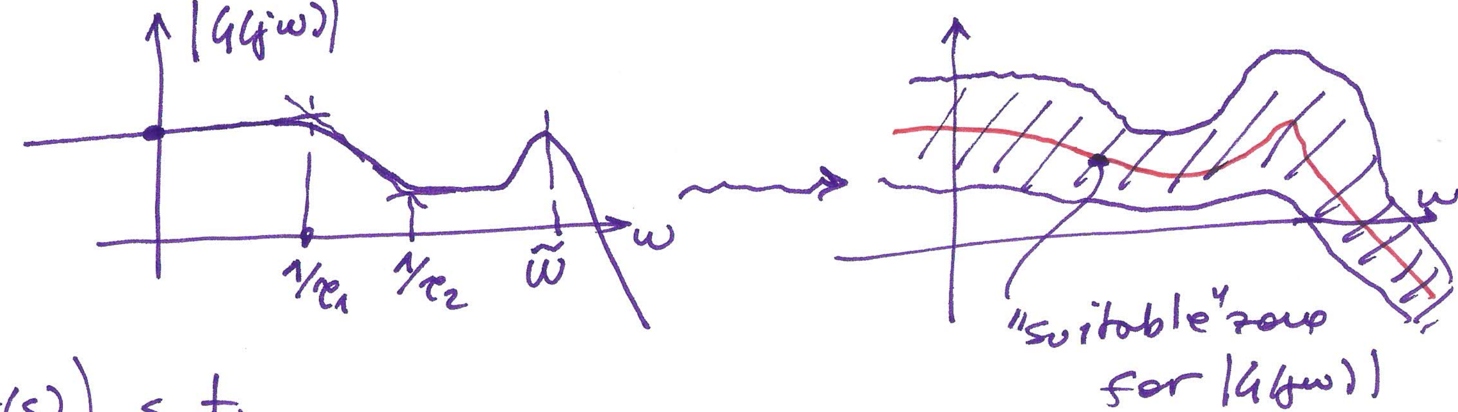
$\Rightarrow$ We can reconstruct by points $G(j\omega)$

Problem: In Aut. Control methods
for stabilizing the system $G(s)$ is used.

i.e. $G(s)$ is assumed to be "perfectly" known.
i.e. the model is already available. ⇒ NOT useful to make experiments.

But: if → (A) $G(s)$ NOT perfectly known. We have some uncertainties
→ (B) We have some Empirical method to stabilize the system (heuristic methods)

(A) ⇒ by using "robust control" methods ⇒ stabilize the system from a "Nominal" transfer function.



⇒ $R(s)$ (and $H(s)$) s.t.
the system is stabilized for any possible
plot inside the //// zone.

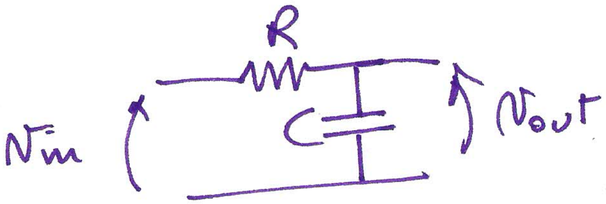
⇒ experiment is useful to refine our knowledge

$\Rightarrow$ we can improve the control scheme after a more more precise model is obtained.

(B) Ex: Ziegler - Nichols method for design of PID.

In "grey-box" case $\Rightarrow$ structure of the model is known but some parameter are not (or not perfectly) known.

ES. RC circuit. $\quad v_{in} \rightarrow \boxed{RC\ circ.} \rightarrow v_{out}$



$$G(s) = \frac{1/RC}{s + 1/RC} = \frac{1}{1 + \underbrace{RC}_{= \tau} s} = \frac{1}{1 + s\tau}$$

the only param. is τ
maybe we have a nominal value for $\tau = RC$ ore we do not have

More generally:

Assume a system has transf. function $\quad G(s) = \dfrac{K}{1 + s\tau}$

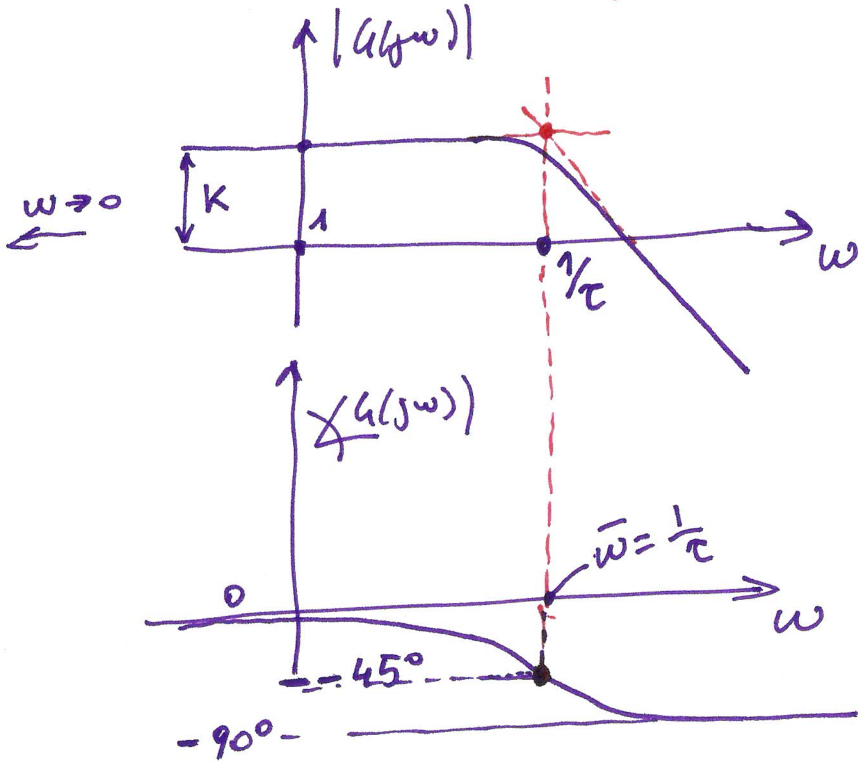
$\Rightarrow$ identification of K and τ (possibly unknown) (I 20)

In cases like this we can use TAYLOR-MADE Methods.

Consider a 1st order system with $G(s) = \frac{K}{1+s\tau}$

$K, \tau > 0$

① The Bode Plot for $G(s) = \frac{K}{1+s\tau}$ is like:



$\Rightarrow$ Experiments for $\xrightarrow{u}$ $\boxed{G(s)}$ $\xrightarrow{y}$

$u(t) = M \sin \omega t$

$\Rightarrow$ change the value of ω until we have

$\measuredangle G(j\omega) = -45°$ $\qquad \bar{\omega}$ s.t. $\measuredangle G(j\bar{\omega}) = -45°$

$$\tau = \frac{1}{\bar{\omega}}$$

Experimentally: $\omega_{\text{temptative}}$.

if $\measuredangle G(j\omega) > -45° \Rightarrow$ augment ω

if $\measuredangle G(j\omega) < -45° \Rightarrow$ reduce ω

until $\measuredangle G(j\omega) = -45° \longrightarrow \bar{\omega}$

Then chose a suitably (some decades) smaller frequency

$\Rightarrow |G(j\omega)| \cong K$.

2 based on <u>step response</u> $\qquad u(t) = 1(t) \longrightarrow \boxed{G(s)} \longrightarrow y(t)$

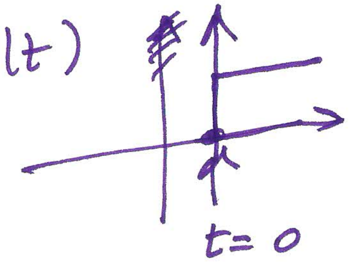
$$G(s) = \frac{K}{1+s\tau} \Rightarrow y_{\text{forced}}(t) = K\left(1 - e^{-t/\tau}\right) \qquad t \geq 0$$

But we can measure $y(t) = y_{\text{free}}(t) + y_{\text{forced}}(t)$

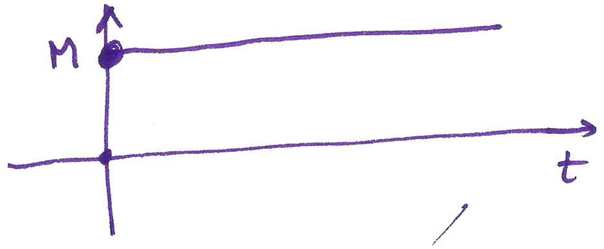
$y_{free}(t)$ due to initial conditions. (zero input response) I22

$y_{forced}(t)$ due to the input (zero initial conditions)

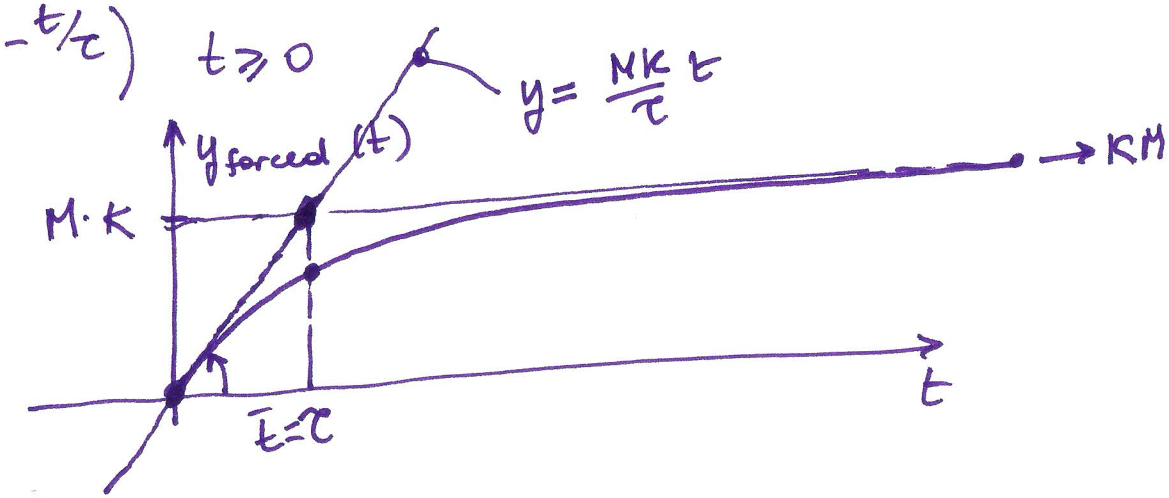
As the system is asymptotically stable ⇒ give $u(t)=0$ for some time until $y(t) \longrightarrow 0$ ⇒ start with $u(t)$



$$\Rightarrow u(t) = 1(t)\cdot M = \begin{cases} 0 & t<0 \\ M & t \geq 0 \end{cases}$$



$$\Rightarrow y_{forced}(t) = M\cdot K\cdot\left(1-e^{-t/\tau}\right) \quad t \geq 0$$



$$y_{forced}(\tau) = M\cdot K\left(1-e^{-1}\right)$$

$$\frac{d}{dt}y_{forced}(0) = \frac{MK}{\tau}$$

⇒ give $u(t) = \cancel{M1(t)}\ M\,1(t)$ for null initial cond.

On the plot of $y(t)$ draw the tangent ~~of~~ in the origin.

$\bar{t}$ s.t. this line value is $M \cdot K$ corresponds to

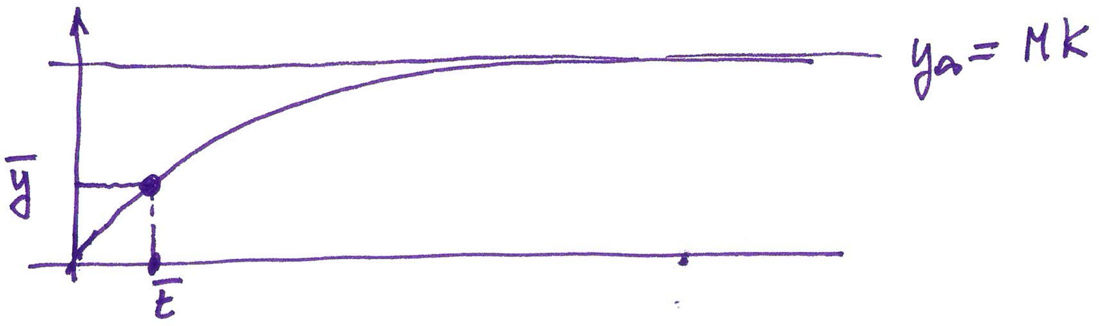
$$\bar{t} = \tau$$

To obtain K we can look at the asymptotic value of $y(t)$

$$y_\infty = MK$$

$$K = \frac{y_\infty}{M}$$

3 for step input $\quad y_{forced}(t) = MK(1 - e^{-t/\tau})$



choose $\bar{t} > 0$ (not too small, not too big)

$\downarrow$ num. errors $\qquad \downarrow$ y not sensitive to different t

$$\bar{y} = MK(1 - e^{-\bar{t}/\tau})$$

solve w.r. to τ

Before doing this: consider the asymptotic value for y and obtain K (as before)

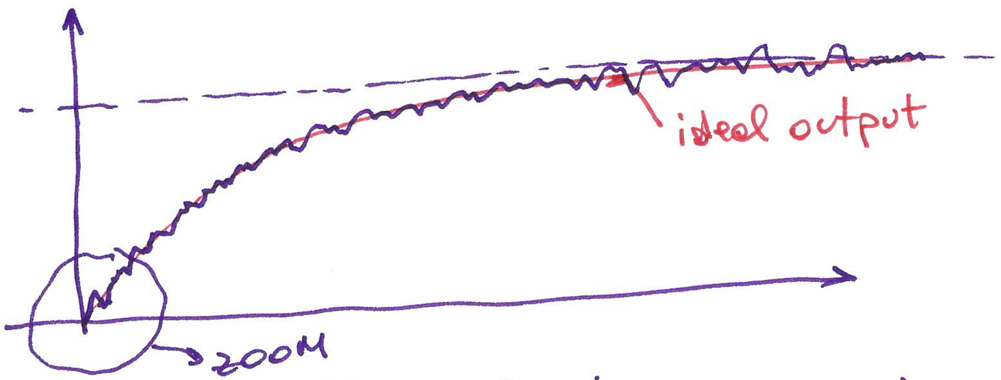
$$\frac{\bar{y}}{MK} = 1 - e^{-\frac{\bar{t}}{\tau}}$$

$$e^{-\bar{t}/\tau} = 1 - \frac{\bar{y}}{MK} \Longleftrightarrow -\frac{\bar{t}}{\tau} = \ln\left(1 - \frac{\bar{y}}{MK}\right)$$

$$\tau = -\frac{\bar{t}}{\ln\left(\underbrace{1 - \frac{\bar{y}}{MK}}_{\text{always} < 1}\right)} \Rightarrow \ln(\cdot) < 0 \Rightarrow \tau > 0$$

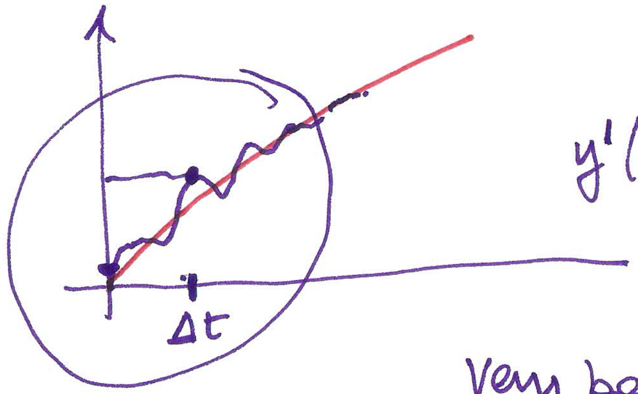
Attention:

When obtaining the plot for $y(t)$ we have noise



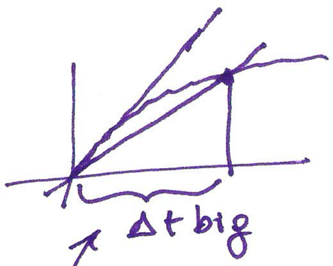
Note: 1) How to compute the ~~sl~~ tang. in $t=0$?

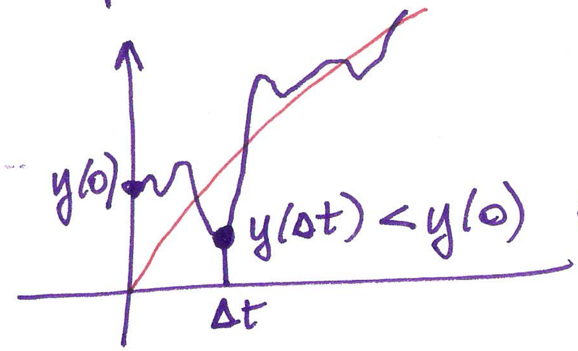
ZOOM:



$$y'(0) \simeq \frac{y(\Delta t) - y(0)}{\Delta t}$$

Very bad possibility:





due to noise.

Δt big $\Rightarrow$ noise is negligible
but bad value for y'

Δt small $\Rightarrow$ noise will be "dominant" w.r.t. Δt
$\Rightarrow$ computation doesn't make sense